Sections: All CS and SE

Probability and Statistics (MT-2005)

Homework #01 (b)

Data Visualization (Qualitative Data)

Question 1

Bar Graph

You are given the following data for the number of students enrolled in different courses:

- Math: 120 students
- Physics: 80 students
- Chemistry: 100 students
- Biology: 90 students

Create a bar graph to show the number of students in each course. The $x - axis$ will represent the course names, and the $y - axis$ will represent the number of students.

Python Task: Use the following Python code to plot the bar graph:

```python
import matplotlib.pyplot as plt

courses = ['Math', 'Physics', 'Chemistry', 'Biology']
students = [120, 80, 100, 90]

plt.bar(courses, students)
plt.xlabel('Courses')
plt.ylabel('Number of Students')
plt.title('Number of Students Enrolled in Different Courses')
plt.show()
```

Question 2

Pie Chart

A company recorded the distribution of its sales by product category for the year:

- Electronics: 45%
- Furniture: 30%
- Clothing: 15%
- Other: 10%

Create a pie chart to represent the sales distribution by category.

Python Task: Use the following Python code to create a pie chart:

```
import matplotlib.pyplot as plt

categories = ['Electronics', 'Furniture', 'Clothing', 'Other']
sales = [45, 30, 15, 10]

plt.pie(sales, labels=categories, autopct='%1.1f%%')
plt.title('Sales Distribution by Product Category')
plt.show()
```

Question 3

Pareto Chart

Concept: A Pareto chart is a bar chart that represents the frequency of events in descending order. It is often used in quality control and decision-making.

Numerical Problem:

Given the following data on the frequency of types of errors in a software application:

Error Type	Frequency
Syntax	50
Logic	30
Runtime	15
Compilation	5

Create a Pareto chart.

Python Task:

1. Plot the Pareto chart showing the frequencies of errors.
2. Include a cumulative percentage line on the chart.

```
import matplotlib.pyplot as plt
import numpy as np

# Data
error_types = ['Syntax', 'Logic', 'Runtime', 'Compilation']
frequencies = [50, 30, 15, 5]

# Calculate cumulative percentages
sorted_freq = sorted(frequencies, reverse=True)
cumulative_percentages = np.cumsum(sorted_freq) /

np.sum(sorted_freq) * 100

# Create a bar plot for the Pareto chart
fig, ax1 = plt.subplots()

ax1.bar(error_types, sorted_freq, color='b')
ax1.set_xlabel('Error Type')
ax1.set_ylabel('Frequency', color='b')

# Create a secondary axis for the cumulative percentage line
ax2 = ax1.twinx()
ax2.plot(error_types, cumulative_percentages, color='r',

marker='o', label='Cumulative Percentage')
ax2.set_ylabel('Cumulative Percentage', color='r')

plt.title('Pareto Chart of Software Errors')
plt.grid(True)
plt.show()
```